Volume & 3D Shapes

Year 4 · **Lesson Worksheet**

Worked Examples

Easy: A cuboid is built from centimetre cubes. It is 3 cubes long, 2 cubes wide and 2 cubes high. How many cubes make it up?

Answer: 3 × 2 × 2 = 12 cubes.

Standard: How many faces, edges and vertices does a cube have?

Answer: A cube has 6 faces, 12 edges and 8 vertices.

Advanced: What is the volume of a cuboid measuring 4cm × 3cm × 2cm?

Answer: Volume = length × width × height = 4 × 3 × 2 = 24 cm^3.

Easy / Standard (15 questions)

1. What is the volume of a cuboid measuring 3cm × 2cm × 2cm?
2. How many faces does a cube have?
3. Which 3D shape has 6 faces, 12 edges and 8 vertices? A) sphere B) cube C) cone D) cylinder
4. How many edges does a cube have?
5. How many vertices does a cube have?
6. What is the volume of a cuboid measuring 4cm × 2cm × 1cm?
7. Which shape has NO flat faces at all? A) cube B) cuboid C) sphere D) cone
8. What is the volume of a cube with sides of 2cm?
9. Select ALL of these 3D shapes that have at least one curved surface: A) sphere B) cube C) cylinder D) cuboid E) cone
10. How many faces does a cuboid have?
11. What is the volume of a cuboid measuring 5cm × 1cm × 3cm?
12. Which shape is a cube an example of? A) a sphere B) a type of cuboid C) a cylinder D) a cone
13. How many edges does a cuboid have?
14. What is the volume of a cuboid measuring 6cm × 1cm × 1cm?
15. How many vertices does a cuboid have?

Standard (15 questions)

1. A cuboid measures 4cm × 3cm × 2cm. What is its volume?
2. Which cuboid has a volume of 24cm^3? A) 4×3×2 B) 6×2×2 C) both A and B D) neither
3. A cuboid measures 5cm × 2cm × 2cm. What is its volume?

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4. How many faces does a triangular prism have?
5. How many vertices does a triangular prism have?
6. Which 3D shape has exactly 2 flat circular faces and 1 curved face? A) sphere B) cylinder C) cone D) cube
7. A cube has a volume of $8cm^3$. What is the length of one side?
8. Select ALL of these cuboids that have a volume of MORE than $20cm^3$: A) 3×3×3 B) 4×2×2 C) 5×2×3 D) 6×1×2 E) 2×2×2
9. How many edges does a square- based pyramid have?
10. How many vertices does a square-based pyramid have?
11. A cuboid measures 3cm × 3cm × 4cm. What is its volume?
12. Which shape has 5 faces, 8 edges and 5 vertices? A) triangular prism B) square-based pyramid C) cube D) cuboid
13. A cube has sides of 3cm. What is its volume?
14. How many faces does a cone have (flat and curved together)?
15. A cuboid measures 2cm × 2cm × 6cm. What is its volume?

Standard / Hard (15 questions)

1. Knowing that a cuboid's volume comes to $24cm^3$, with a length of 4cm and a width of 3cm, work out its height.
2. Picture a box with dimensions 3cm × 2cm × 4cm, completely packed with 1cm cubes. Select ALL of the statements below that are TRUE: A) the box holds 24 small cubes B) the box's volume is $24cm^3$ C) the box holds 9 small cubes D) each small cube has a volume of $1cm^3$ E) the box's volume is $9cm^3$
3. Comparing two cuboids — Cuboid A at 4×2×3cm and Cuboid B at 3×4×2cm — which one takes up more volume?
4. Given that a cuboid has a volume of $40cm^3$, a length of 5cm and a height of 4cm, find its width.
5. With a volume of exactly $8cm^3$, a cube-shaped box must have edges of what length? A) 2cm B) 4cm C) 6cm D) 8cm
6. Joined end to end in a single row, four identical 1cm cubes form one long cuboid. State its dimensions, and work out its total volume.
7. A fish tank measures 8cm × 5cm × 2cm. What is its volume? A) $60cm^3$ B) $70cm^3$ C) $80cm^3$ D) $90cm^3$
8. A reference cuboid measures 6cm×2cm×2cm. Which of the cuboids listed below shares that exact same volume? A) a 4cm×3cm×2cm cuboid B) a 3cm×4cm×1cm cuboid C) both A and B D) neither
9. A cube's single face has an area of $9cm^2$. What is the cube's volume?
10. A cuboid measures 4cm × 4cm × 2cm. Is this a cube? Explain your answer, and give its volume.
11. A cuboid has a square base of side 3cm and a height of 4cm. What is its volume?
12. A cuboid measures 4cm × 2cm × 3cm. If the length is doubled to 8cm, what is the new volume?
13. Select ALL of the cuboids below that could be built from exactly 12 unit cubes: A) 4×3×1 B) 6×2×1 C) 3×2×2 D) 4×2×2 E) 2×2×3
14. A box measuring 5cm×2cm×2cm is completely filled with 1cm cubes, then half of them are removed. How many cubes are left?
15. A cuboid box is made from edges of 5cm, 3cm and 2cm. What is the volume of the box?

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